



Green University of Bangladesh
Department of Computer Science and Engineering
Faculty of Sciences and Engineering
Semester:(Fall, Year: 2024), B.Sc. in CSE (Day)

Portfolio

Course Title: Microprocessor and Microcontroller
Course Code: CSE 303, Section: 222 D6

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Submission Date : 25.12.2024

Assignment Status

Marks:
Comments:.....

Signature:.....
Date:.....

8086

- ⇒ 20-bit address line
- ⇒ 16-bit data lines
- ⇒ 16-bit microprocessor.
- ⇒ Two modes of operation
 - ↳ Min
 - ↳ Max

Flag register: 16 bit register that can hold 16 flip-flops. 9 bits are used as flag.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U	U	U	U	OF	DF	IF	TF	SF	ZF	U	AF	U	PF	U	CF

U = Undefined

6 = status flag

3 = Control flag

$x \geq y = CF = 0$

even, PF = 1

odd, PF = 0

result = 0, Z = 1

S = 1, result negative set.

Addition:

$$\begin{array}{r}
 \begin{array}{cccc}
 0110 & 0101 & 1101 & 0001 \\
 0010 & 0011 & 0101 & 1001 \\
 \hline
 1000 & 1001 & 0010 & 1010
 \end{array}
 \end{array}$$

↑ 8bit

SF=1 CF=0 AF=1 PF=1 ZF=0 OF=0

Z=0 result $\neq$ 0

	S	PF	AF	CF	ZF
Set	1	1	1	1	1
Result	0	0	0	0	0

IF = 1 = INTR enabled

Interrupt Vector Table (IVT)

256 Interrupts range 0000H - 03FFH

Each Interr holds = 4 byte address

IVT requires 4 byte X 256

= 1 KB memory.

For type 64,

type 0
type 1

$$\text{Starting address} = 4 \times 64$$

$$= 256$$

$$= 100H$$

$$= 00100H \leftarrow 20 \text{ bits address}$$

4 bytes of each Interrupt types holds data CS and IP address. Total ISR address = $CS \times 10H + IP$

INT N,

N = Type Number = 0 to 255

1KB is located in the starting part of memory

$$CS : IP = 0000 : 0000 \text{ to } 0000 : 003FF H$$

IP for the ISR of Type N = 4N

CS " " " " " N = 4N + 2

for type 1, IP = $4 \times 1 = 00004H$

$$CS = 4 \times 1 + 2 = 00006H$$

Example: Determine the physical address of the ISR for the given IVT if Type 0.

Type 0, $N = 0$

$$IP, 4N = 4 \times 0 = 00000H$$

$$CS, 4 \times 0 + 2 = 00002H$$

$$IP \text{ of the ISR} = 3405H$$

$$CS \text{ of the ISR} = 2000H$$

$$\text{Physical address} = CS \times 10H + IP$$

$$= 2000 \times 10 + 3405$$

$$= 23405H$$

(A)

18.11.24

Example:Answer:Type 255, $N = 255$

$$IP = 4N = 4 \times 255 = 1020D = 003FCH$$

(Address of ISR)
IP

IP is stored at this address. This is not the value of the IP, it is the value at which the IP of the ISR will be located.

$$CS = 4N + 2 = 4 \times 255 + 2 = 1022D = 003FEH$$

(Address of the ISR CS)

$$IP \text{ of the ISR} = 3322H$$

$$CS \text{ of the ISR} = 4000H$$

$$\text{Physical Address} = CS \times 10H + IP$$

$$= 3322 \times 10 + 4000$$

$$= 43322H$$

(Ans)

~~The difference between minimum mode and maximum mode:~~

Starting address of type 28:

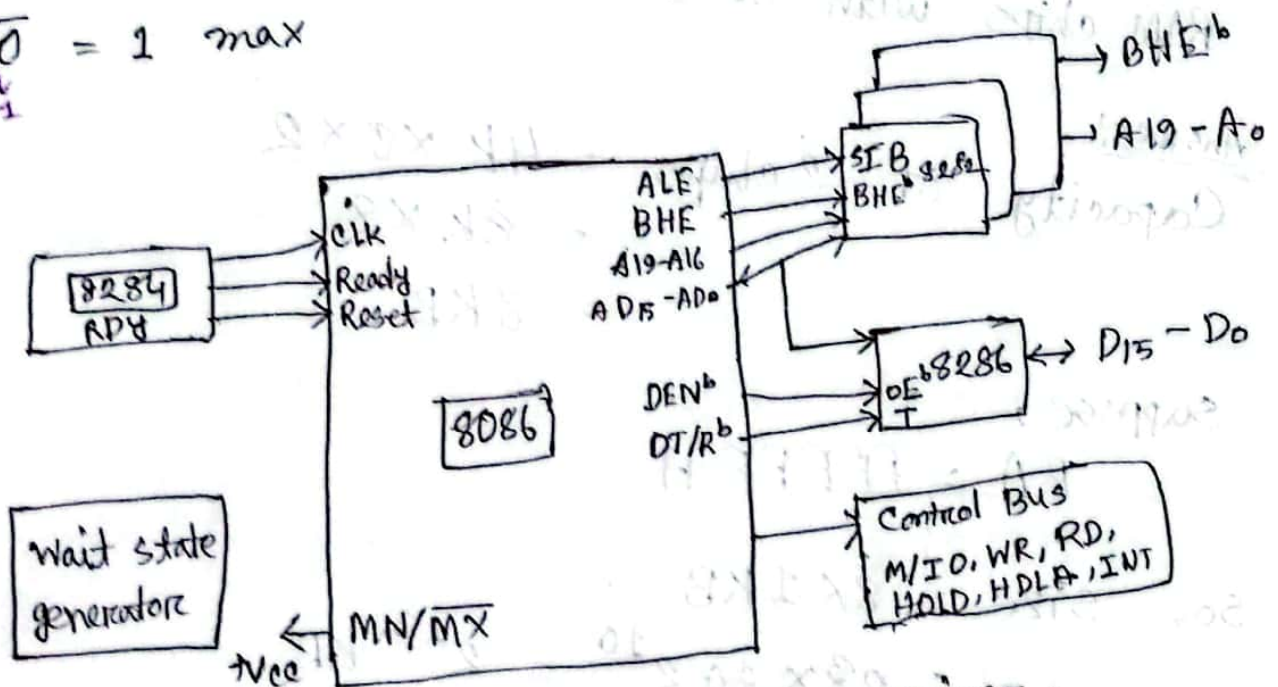
$$SA = \text{INtr type} \times 4$$

$$= 28 \times 4$$

$$= 112 \text{ decimal} = 70 \text{ hex} = 0070H$$

$$\overset{1}{\text{MN}} / \overset{1}{\overline{\text{MX}}} = 1 = \text{minimum mode}$$

$$\text{Or } \underset{0}{0} / \underset{1}{\overline{0}} = 1 \text{ max}$$



$\overline{\text{DEN}}$, $\text{DT}/\overline{\text{R}}$ are used for transreceiver.
Two transreceivers are needed.

$\overline{\text{DEN}}$	$\text{DT}/\overline{\text{R}}$	Action
1	X	Transreceiver is disabled.
0	0	Receive Data.
0	1	Transmit data.

Qs0 and Qs1 tells the status of the queue.
 If 0 = Qs1 and 0 = Qs0, no operation in queue.
 S2, S1 and S0 together tells the status of CPU cycle.

Memory Interfacing:

8086 = 20bit address line so $2^{20} = 1\text{MB}$ memory

Memory range 00000F - FFFFFFF H

Interface two 4K x 8 EPROM and Two 4K x 8 RAM chips with 8086.

Answer:

Capacity of 2 chips = $4\text{K} \times 8 \times 2$
 $= 8\text{K} \times 8$
 $= 8\text{KB}$

Suppose,

EA = FFFFFFF H

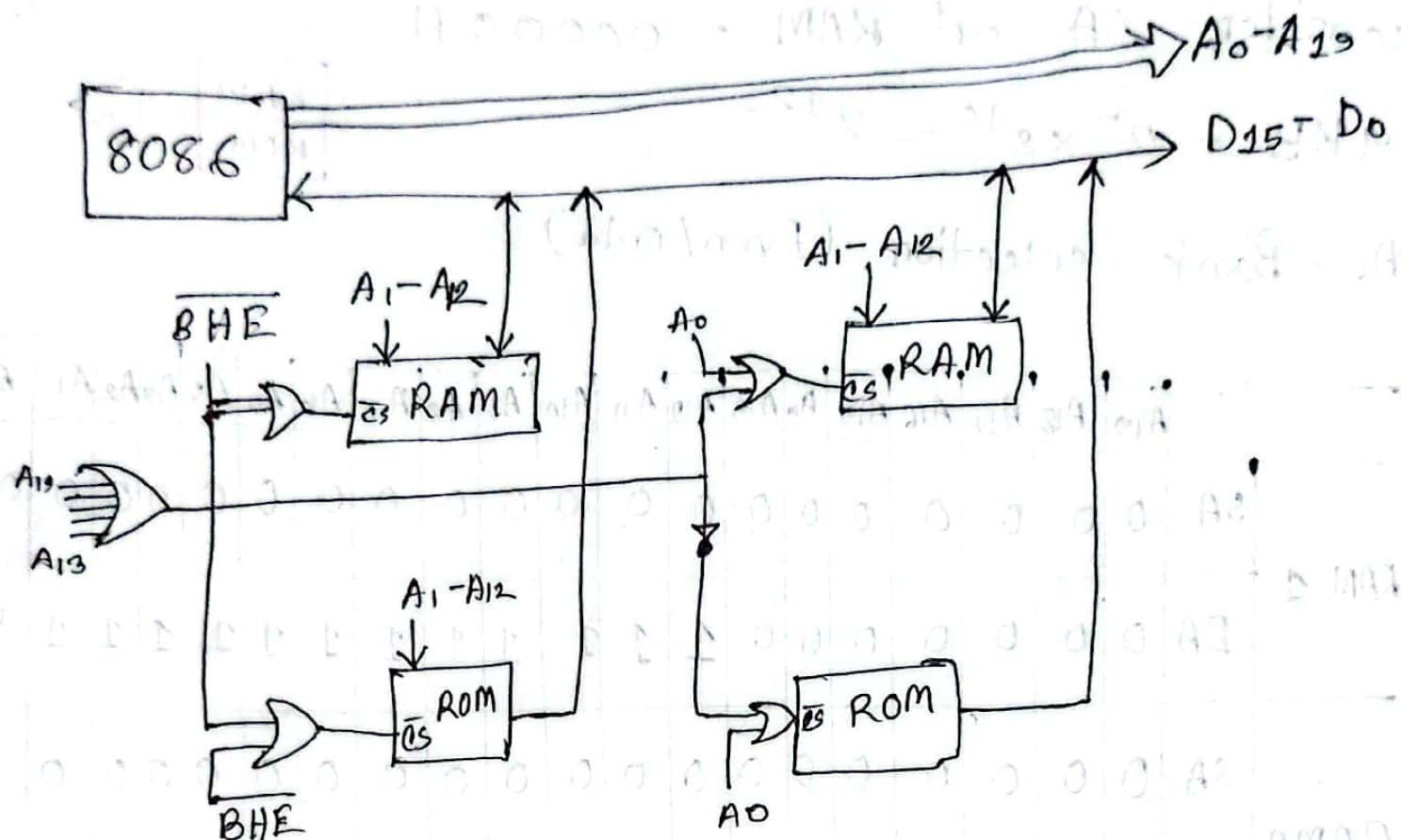
So, Size = $8 \times 1\text{KB}$

$$= 2^3 \times 10^3 = 2^{13} \text{ KB}$$

0000 0001 1111 1111 1111 = 01FFF H

SA of ROM = FFFFFFF - 01FFF H

$$= \text{FE000H}$$



There are two types of interfacing:

① Memory interfacing

② I/O interfacing

Memory interfacing
 address line 20 bits
 8 bit data line @ 16 bit

I/O interfacing

8 bit Data line +
 8 bit address line

8086 memory Organization:

Used distinguish between odd and even bytes

The CPU provide signal $\Rightarrow$ BHE

BHE ও A_0 ব্যবহার করা হয় odd এবং even bank select করার জন্য।

BHE	A_0	Function
0	0	odd + even
0	1	only odd
1	0	only even
1	1	None

I/O device ২ভাবে interface করা যায়:

- ① Isolated mapped I/O
- ② Memory mapped I/O

Have to read!
Advantage + disadvantage

EPROM এর জন্য IC $\Rightarrow$ 27xxx

এবং RAM IC $\Rightarrow$ 61xxx

৯৫,

IC 27128 ইয়া আইসন EPROM

Memory size 128 KB

$$\Rightarrow \frac{128}{8}$$

$$\Rightarrow 16 \text{ KB}$$

$$= 2^4 \times 2^{10} \cdot 8$$

$$= 2^{14} \cdot 8$$

$m=14$, $n=8$ ← data line
↑
address line

☐ Logically = 1Mx8 memory bank

Physically $\Rightarrow$ 512 KBytes $\Rightarrow$ The even bank
low
and the odd bank.
high

For even bank $A_0 = 0$

For odd bank $A_0 = 1$

Interface micro 8086 with two of IC —
27512 chip.

Memory size = 512×8
= 64KB

= $2^{16} \times 8$

Address line = 16

Data line = 8

Microcontroller

- Small in size.
- built in RAM, ROM, I/O, up, serial ports, timers, clock circuit.
- On-chip computer.
- CMOS technology is used.

Types of microcontroller:
Based on bits → 8bit, 16bit, 32bit

- External and Internal.
- sets: RISC, CISC.
- Microcontroller has 8bit data bus and 16 bit address bus [8051]

→ 8051 has 4 Ports → drivers
↑
Buffer & latch are there

Characteristics of 8051:

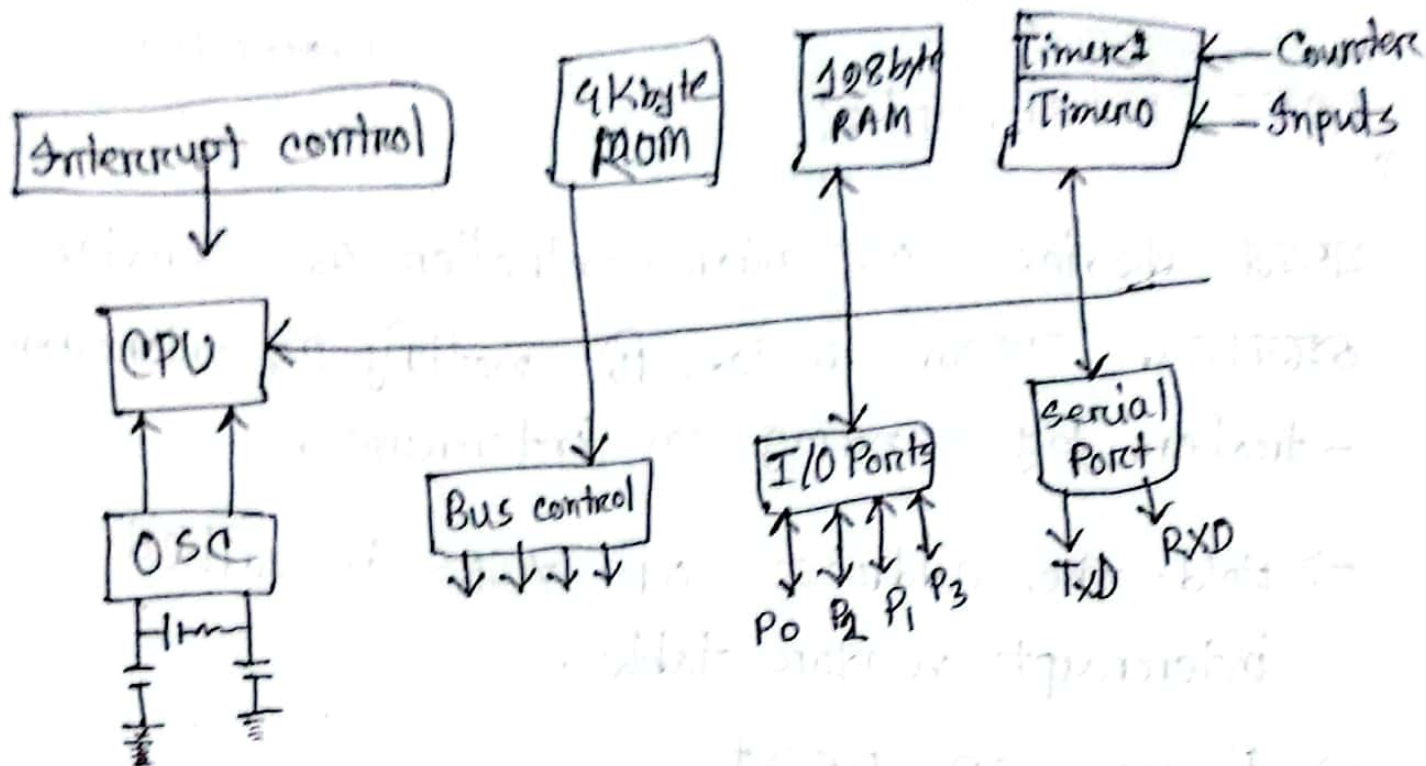
- ① 8 bit ports [Bidirectional]
- ② It has registers, RAM, and ROM
- ③ ROM → There is no address register
- ④ RAM → There is address register, PSEN, SP, ALU, PSEN, ALE, Oscillator.

 CPU is the brain of any processing device of microcontroller.

8051 microcontroller interrupts →

Various types → Maskable, non-maskable hardware, software mainly 5.

- Interrupts →
- ① INTO
 - ② TFO
 - ③ INT1
 - ④ TF1
 - ⑤ R1/T1



Timer and counters ⇒ Two

T₀: Timer 0 external input → P3.4

T₁: Timer 1 external input → P3.5

Gate	ClT	M ₁	M ₀	Gate	ClT	M ₁	M ₀

1: Counter, 0: Timer: Delay is generated.
M₁ and M₀ decides the mode of a operation in timer.

TMOD = 00000

0010

Timer 1 is not used.
Timer 0, mode 2.

C/T = 0 to use XTAL, clock source

Gate = 0 to use internal.

8051 interrupts:

যখন device এর microcontroller এর service প্রয়োজন, তখন device টি notify হয় microcontroller by sending an interrupt.

→ Hold the address of 7 SRS is called interrupt vector table.

→ Power on reset.

→ Two external interrupt.

→ Two timer interrupt.

→ serial Port interrupt.

Peripheral device $\xrightarrow{\text{request}}$ Microcontroller

acknowledgement $\leftarrow$ INTA

$\xrightarrow{\text{send the vector address.}}$

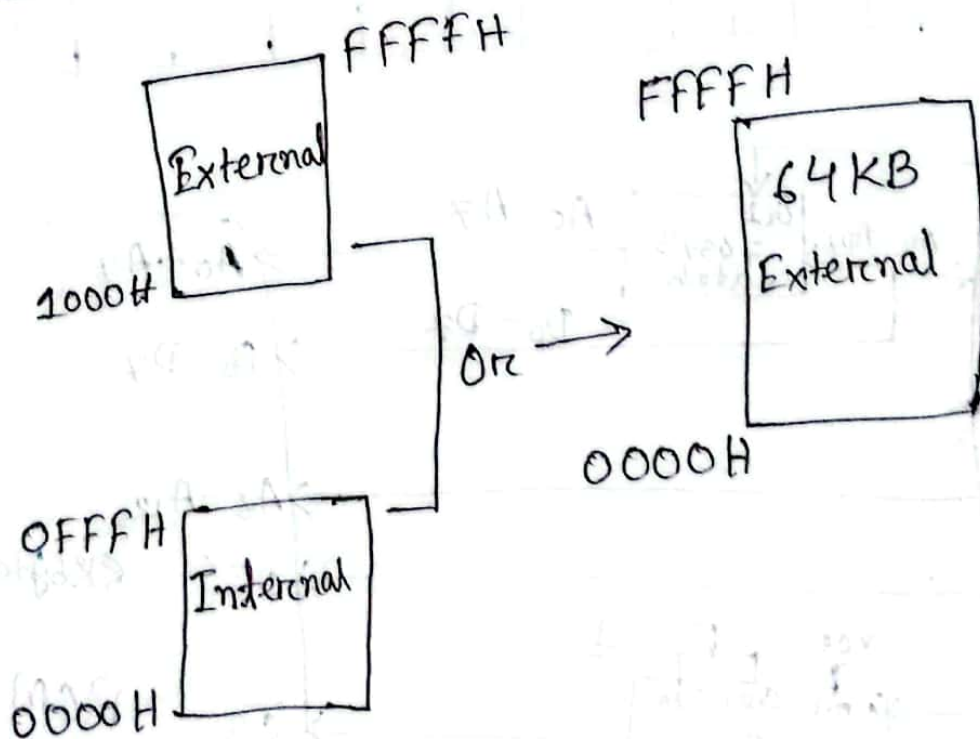
lower nibble of TCON register used with INTO and INT1.

TF flag in TCON register is used

↳ Timer 0/1 interruption

External interrupt	0	↑ highest lowest priority →
Timer interrupt	0	
External interrupt	1	
Timer interrupt	1	
serial communication		

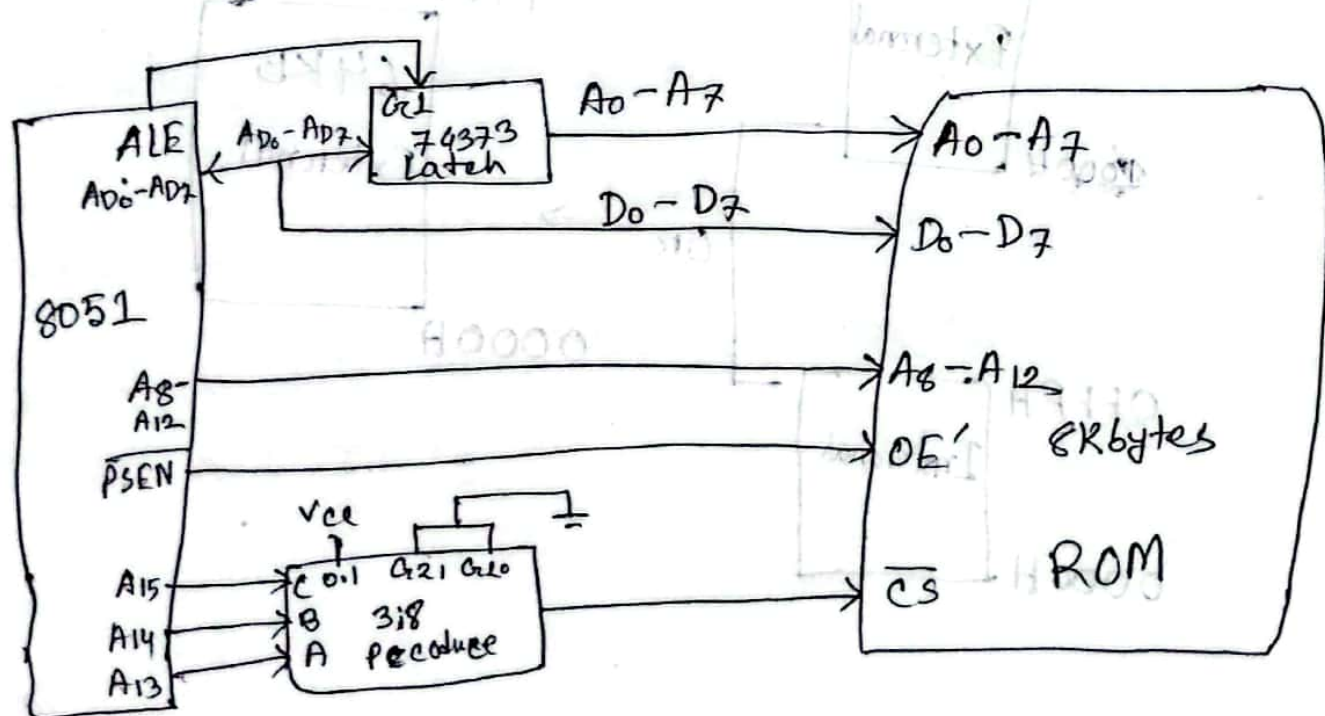
8051 ⇒ Memory Interfacing



* 8KB ROM to 8051 interfacing:

13 address lines
memory map => Data memory mapping

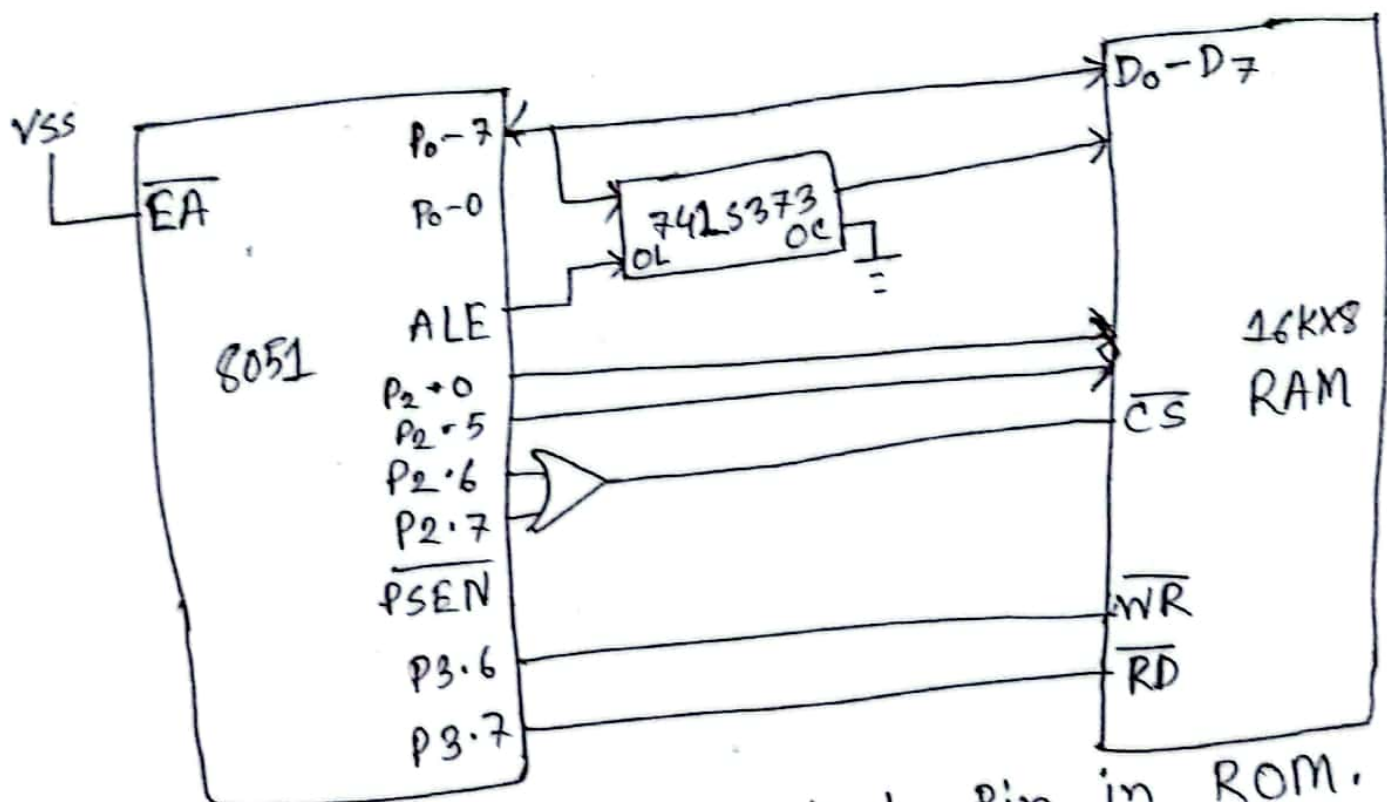
	A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	Add
SA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0000H
2nd Loe	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0001H
Ending	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1FFFFH



Ex - 2 $\Rightarrow$ External

RAM size is $16K \times 8$
14 address lines.

	A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
SF	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	000c H
EA	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	3FFF H



$\overline{PSEN}$ is used to chip select pin in ROM.
 $\overline{RD}$ is used to read control signal pin.
 $\overline{WR}$ is used to write control signal pin.

Microprocessor

8086 :

⇒ 20-bit address line

⇒ 16-bit data line

⇒ 16-bit microprocessor

⇒ Two modes of operation

1. Min 2. Max

Flag register: 16 bit register that can

hold 16 flip-flops 9 bits are used as

Flag.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U	V	U	V	OF	DF	IF	TF	SF	ZF	U	AF	U	PF	U	CF

U is Undefined

6 = status flag

3 = Control flag

$u \geq y = CF = 0$

even, $PF = 1$

odd, $PF = 0$

result, $= 0, Z = 1$

$S = 1$ result negative set.

Addition:

0110	0101	1101	0001
0010	0011	0101	1001
1000	1001	0010	0110

↑ 8bit

$SF = 1$ $CF = 0$ $AF = 1$ $PF = 1$ $ZF = 0$ $OF = 0$

$Z = 0$ result $\neq 0$

	S	PF	AF	CF	ZF
Set	1	1	1	1	1
Result	0	0	0	0	0

Here, $IF = 1$ ~~so~~ $= INTR$ enabled

Interrupt Vector Table (IVT)

256 Interrupts range $0000H - 03FFH$

Each Interr holds ≈ 4 bytes address

IVT requires $4 \text{ byte} \times 256$

$\approx 1 \text{ kB memory}$

For type 69,

Starting address $\approx 4 \times 69$

≈ 276

$\approx 100H$

$\therefore 00100H \leftarrow 20$ bits, address

4 bytes of each Interrupt types holds

data CS and IP address - Total ISR

$$\text{address} = CS \times 10H + IP$$

INT N,

N = Type Number = 0 to 255

1KB is located in the starting part
of memory

$$CS : IP = 0000 : 0000 \text{ to } 0000 : 003FF$$

IP for the ISR of type $N = 4N$

$$CS \quad u \quad u \quad u \quad u \quad u \quad N = 4N + 2$$

for type 1, $IP = 4 \times 1 = 00004H$

$$CS = 4 \times 1 + 2 = 00006H.$$

Example: Determine the physical address of the ISR for the given IVT if

Type 0.

Type 0, $N=0$

$$IP, 4N = 4 \times 0 = 00000H$$

$$CS, 4 \times 0 + 2 = 00002H$$

$$IP \text{ of the ISR} = 3905H.$$

$$CS \text{ of the ISR} = 2000H$$

Physical address: $CS \times 10H + IP$

$$= 2000 \times 10 + 3905$$

$$= 23905H.$$

Microprocessor:

Example:

$$\text{Type } 255, N = 255$$

$$IP = 4N = 4 \times 255 = 1020D = 003FCH$$

(address of ISR IP)

IP is stored at this address. This is not the value of the IP. it is the value at which the IP of the ISR will be located.

$$OS = 4N + 2 = 4 \times 255 + 2$$

$$= 1022D$$

$$= 003FEH \text{ €}$$

$$IP \text{ of the ISR} = 3322H$$

$$OS \text{ of the ISR} = 4000H$$

$$\text{Physical Address} = \text{CS} \times 104 + \text{IP}$$

$$= 3322 \times 104 + 4000$$

$$= 433224$$

* Starting address of type 28.

$$\text{SA} = \text{INTR type} \times 4$$

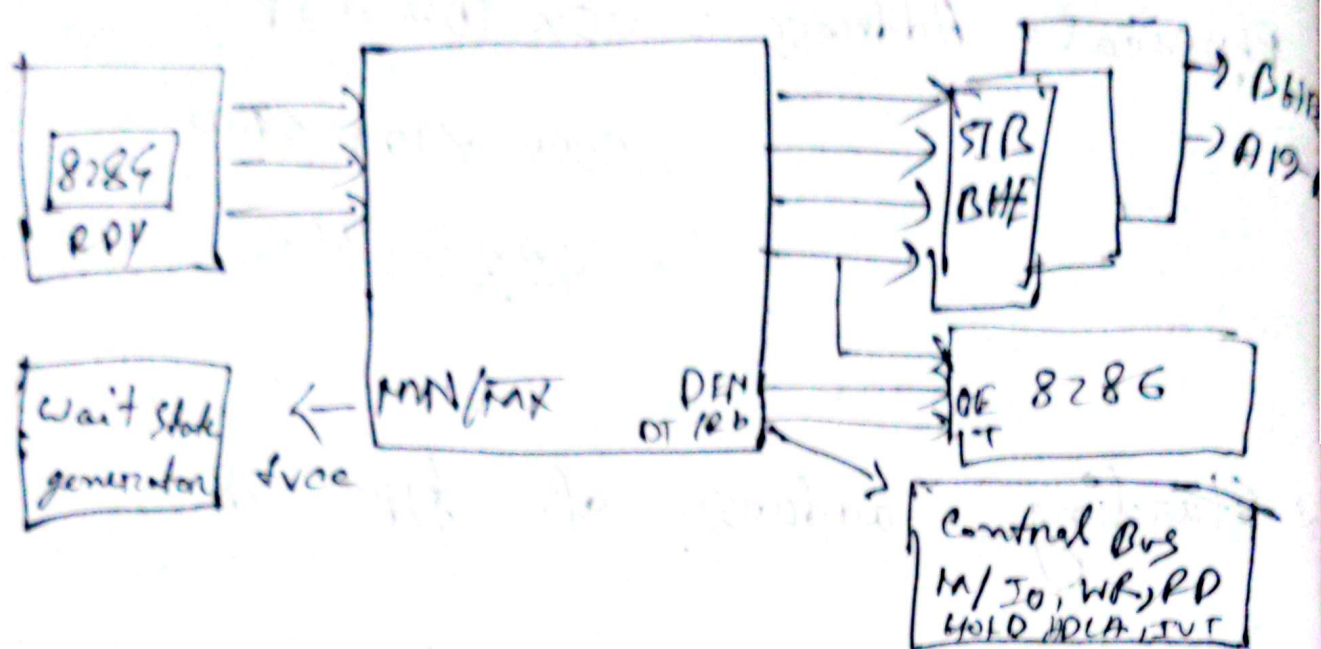
$$= 28 \times 4$$

$$= 112 \text{ decimal}$$

$$= 70 \text{ Hex} = 0070\text{H}.$$

min/max 2 is minimum mode

$$0/0 = 1 \text{ max}$$



$\overline{DEN}$, $DT/\overline{R}$ are used for transceiver
Two transceivers are needed.

DATA $\overline{DEN}$	$DT/\overline{R}$	Action
1	X	Transceiver is disabled
0	0	Receive data
0	1	Transmit data

Q_{S0} and Q_{S1} tells the status of the queue. If $Q_{S1} = 0$ and $Q_{S0} = 1$, No operation is queue S_2, S_1 and S_0 together tells the status of CPU cycle.

Memory Interfacing:

8086 = 20 bit address line so $2^{20} = 1\text{MB}$

Memory range 00000H - FFFFFH.

Microprocessor:

Interface two 4K x 8 EPROM and
Two 4K x 8 RAM chips with 8085

Solution:

$$\text{Capacity of 2 chips} = 4K \times 8 \times 2 \\ = 8KB$$

Suppose,

$$EA = FFFFFFH$$

$$\therefore \text{Size} = 8 \times 1KB$$

$$= 2^3 \times 10$$

$$= 2^{13} KB$$

$$0000 \quad 0001 \quad 1111 \quad 1111 = 01FFFFH$$

$$\text{SA of ROM} = FFFFFF - 01FFFFH$$

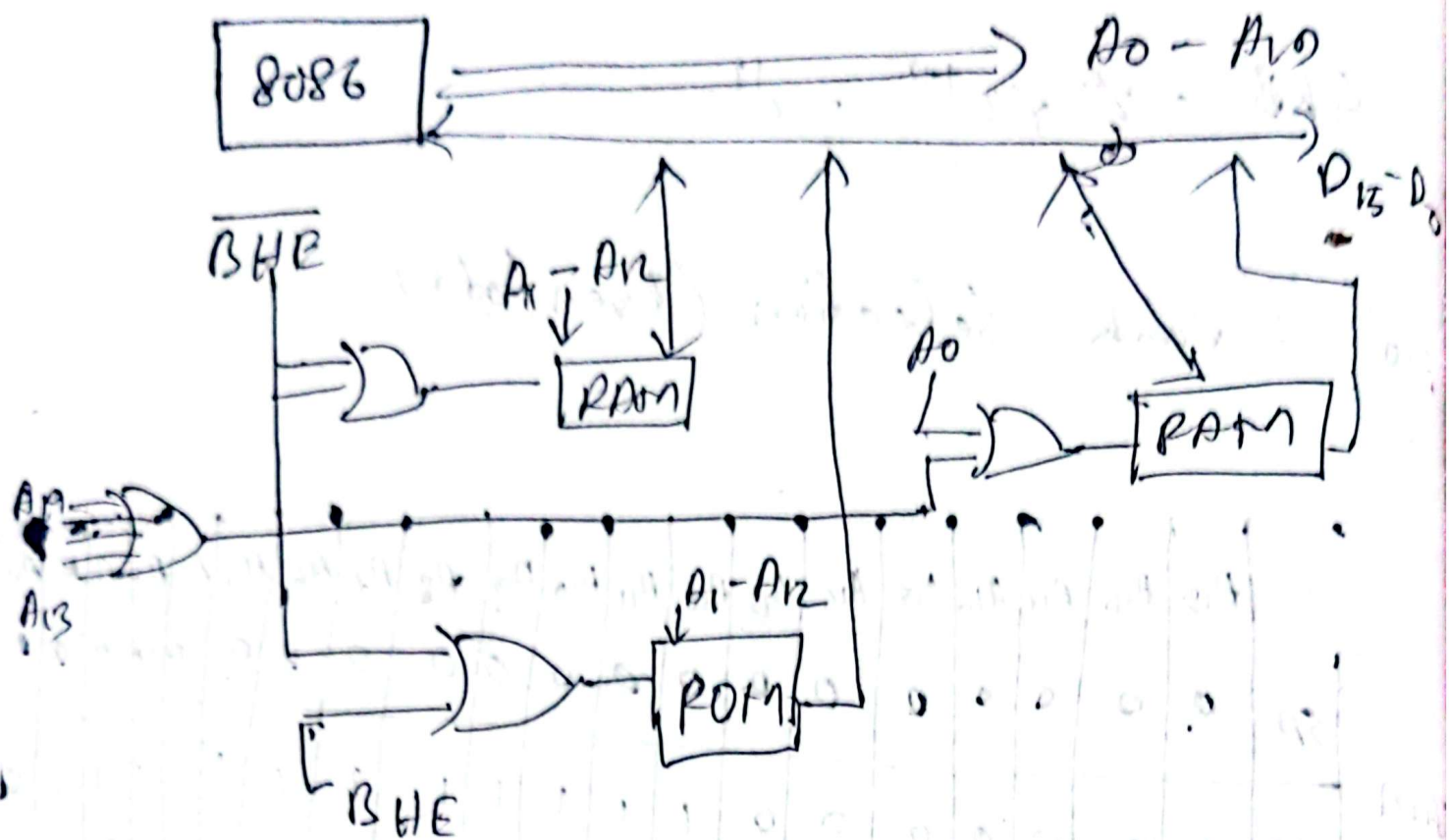
$$= FE0000H$$

consider SA of RAM = 00000 b

$$4\text{KB} = 2^2 \times 2^{10} = 2^{12}$$

A_0 = Bank selection (Even, odd)

[illegible]



There are two types of Interfacing

- ① Memory interfacing
- ② I/O interfacing

Memory interfacing → 8 bit data line

16 bit address line.

I/O interfacing → 8 bit Data line & 8 bit address line

8086 Memory Organization

Used distinguish between odd and even bytes the cpu provide signal $\Rightarrow$ BHE

BHE	A0	Function
0	0	odd + even
0	1	only odd
1	0	only even
1	1	None

Here, I/O device 2 no interface

- ① Isolated mapped I/O
 - ② Memory mapped I/O
- Both have to read to:
Advantage + disadvantage.

EPROM $\Rightarrow$ IC $\Rightarrow$ 27

RAM IC $\Rightarrow$ 61

ff,

IC 27128 - 20 pin - EPROM

Memory Size 128 KB

$$\Rightarrow \frac{128}{8}$$

$$\Rightarrow 16 \text{ KB}$$

$$\Rightarrow 2^9 \times 2^{10} \cdot 8$$

$$\Rightarrow 2^{14} \cdot 8$$

$$m = 2^4, n = 8$$



address line



data line

Microprocessor:

- Small in size
- built in RAM, ROM; I/O Up Serial ports timers. clock circuit.
- On-chip computer
- CMOS technology is used

* Types of Microprocessor:

Based on bits → 8 bits, 16 bits, 32 bit.

- ⇒ External and Internal.
- ⇒ Sets: RISC, CISC
- ⇒ Microcontroller has 8bit bus and 16 bit address bus [8051]

⇒ 8051 has 4 ports → drivers. ^{→ Buffer 8} ^{one then}

Characteristics of 8051:

① 8-bit Ports

② It has register, RAM and ROM

③ ROM → There is no address register.

④ RAM → There is address register

PSN, SP, ALU, PSEN, ALE,

Oscillator.

☐ CPU is the brain of any

processing device of microcontroller

8051 microcontroller interrupts $\rightarrow$

Various types $\rightarrow$ Maskable and

non maskable, hardware, Software
mainly 5.

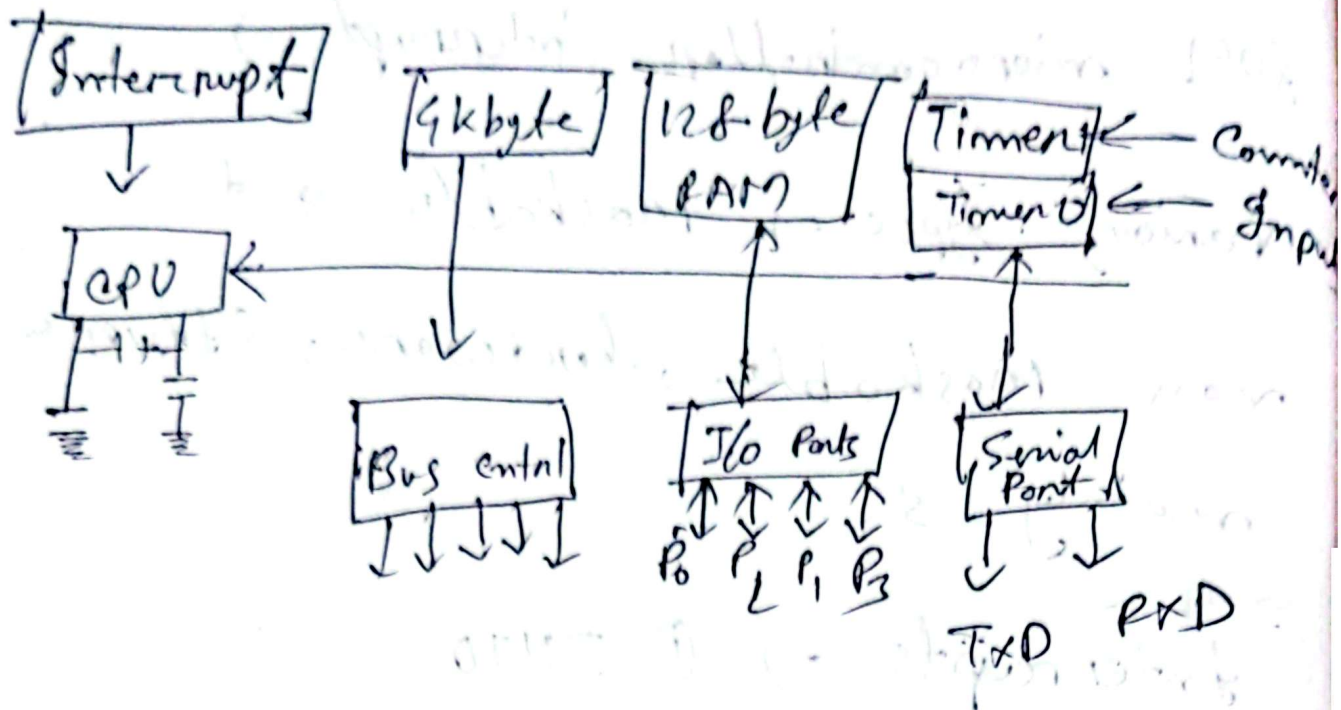
Interrupts $\rightarrow$ ① INTO

② TFO

③ INT₁

④ TF₁

⑤ RI/T₁



Timers and counters:

Two

T_0 : Timer 0 external input $\rightarrow$ P3.4

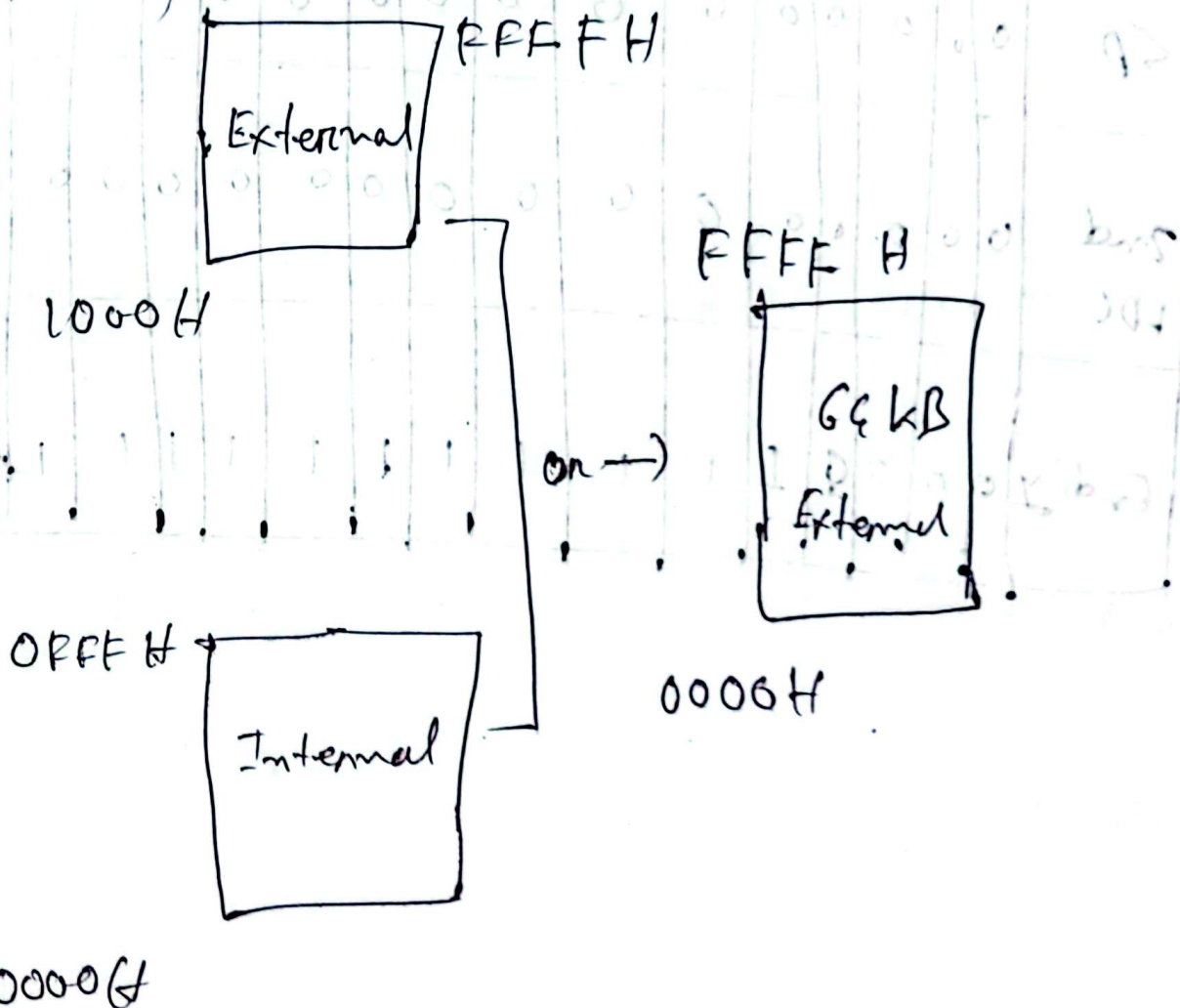
T_1 : " " " " $\rightarrow$ P3.5

Gate	CR	M ₁ M ₀	Gate	C ₁ T	M ₁	M ₀
	Timer 1			Timer 0		

1: Counter 0: Timer: Delay is generated
 M_1 and M_0 decides the mode
 of a operation in timer.

$\text{TRAP} = 0000010010$
 $\uparrow \quad \downarrow$
 $\text{C/T} = 0$ to use XTAL
 clk source
 $\text{Rate} = 0$ to use
 internal.

Q5) ⇒ Memory Interfacing :

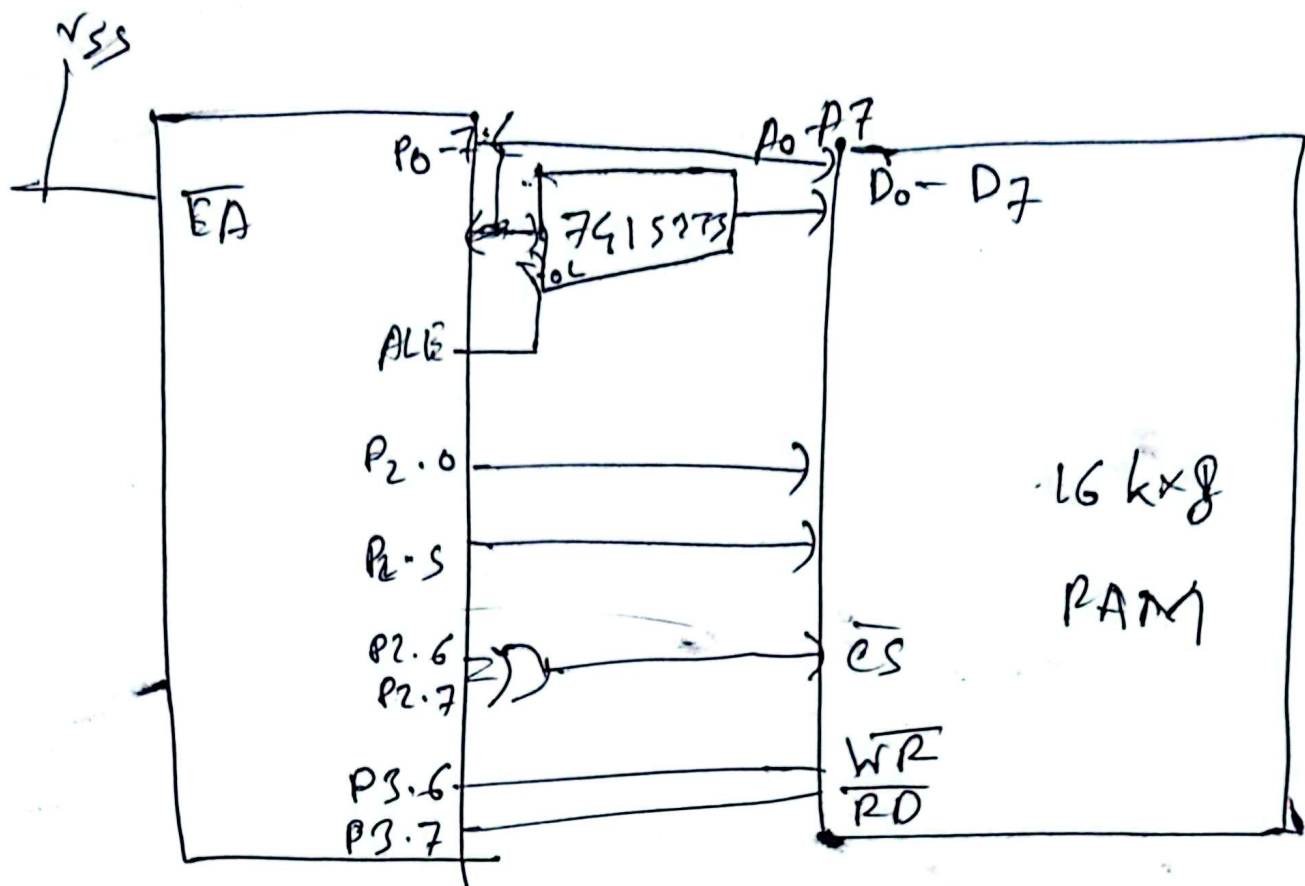


1970-1971

13 address lines

• *Leptocarpus* *Leptocarpus*

10



$\overline{PSEN}$ is used to Chip Select Pin
in ROM.

$\overline{RD}$ is a read signal pin

$\overline{WR}$ is a write control pin.